TP03 - Interactive Data Analysis and Visualization with Power BI

Module: Big Data Analytics | Master 2 AI & Data Science | Student: Djoumana

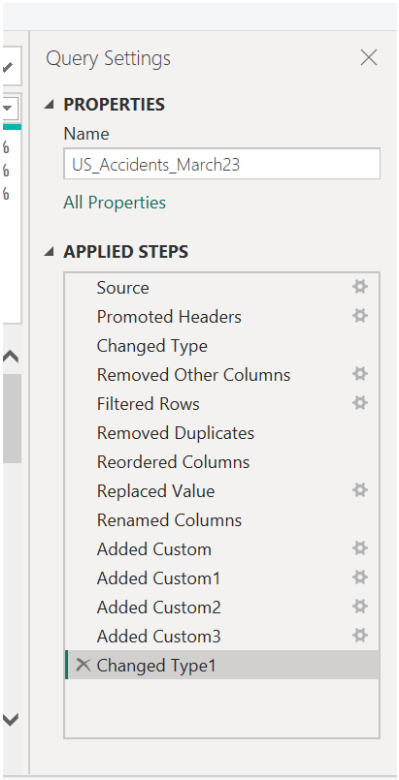
1. Dataset and scope

- **Dataset:** US Accidents (2016 - 2023), Kaggle: https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents
- **Original size:** 7,728,394 rows and 46 columns (CSV file of several GB).
- **Scope used:** California only (State = CA) and 12 selected columns, because of hardware limits. The dashboard therefore describes California, not the whole United States.

2. Import and cleaning (Tasks 1 and 2)

The CSV was imported with Get Data (Text/CSV) and inspected in Power Query: column types, row count, and missing or duplicated values. The following transformations were then applied, as listed in the Applied Steps pane:

- **Choose columns:** kept ID, Severity, Start_Time, Start_Lat, Start_Lng, City, State, Temperature(F), Humidity(%), Visibility(mi), Weather_Condition, Sunrise_Sunset.
- **Filter rows:** State = CA.
- **Remove errors:** one row with a conversion error in Weather_Condition and Visibility(mi) was removed.
- **Remove duplicates** on ID, the unique accident identifier.
- **Missing values:** missing text values (City, Weather_Condition) replaced by "Unknown". Missing numeric values (temperature, humidity, visibility) were left as null so that averages are not biased by artificial zeros.
- **Rename columns:** Start_Time to Accident Start, Start_Lat to Latitude, Start_Lng to Longitude, Weather_Condition to Weather, and so on.
- **Added columns** derived from Accident Start: Year, Month, Hour, YearMonth (4 new columns, 16 in total).



Applied Steps in Power Query

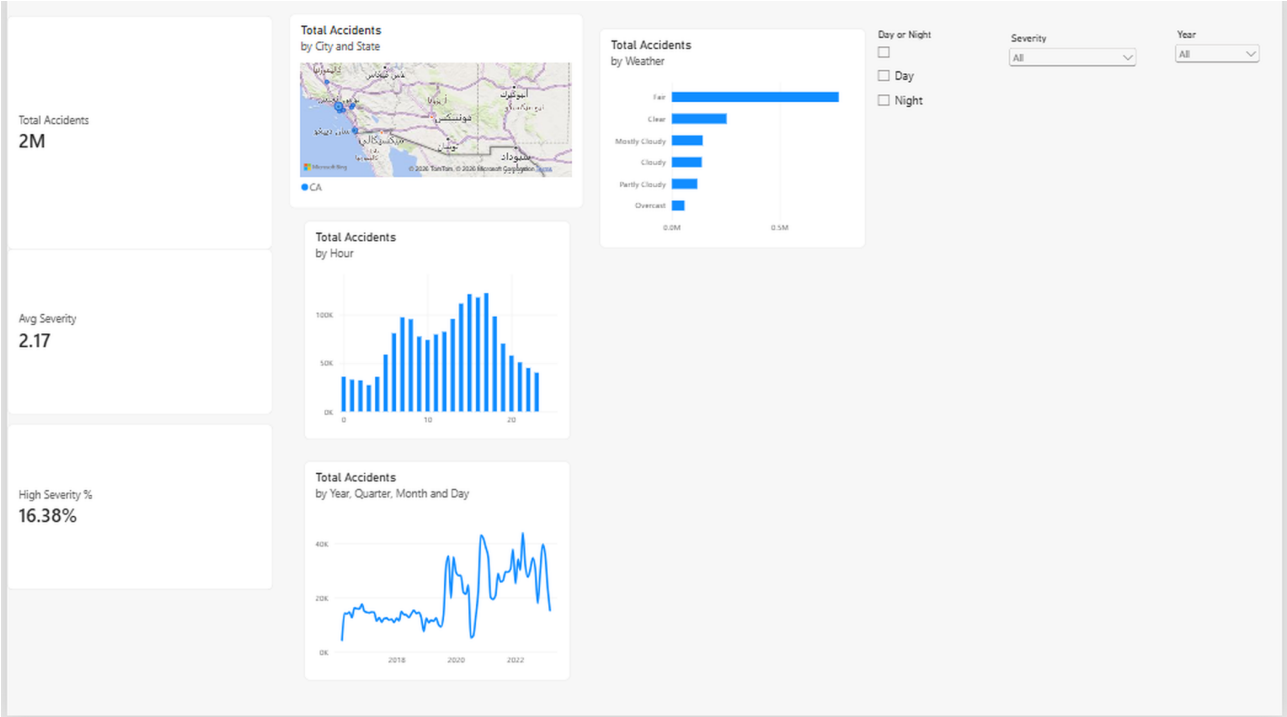
3. DAX measures

Three measures were created and used in the KPI cards:

```
Total Accidents = COUNTROWS(US_Accidents_March23)
Avg Severity = AVERAGE(US_Accidents_March23[Severity])
High Severity % = DIVIDE(CALCULATE([Total Accidents], US_Accidents_March23[Severity] >= 3),
[Total Accidents])
```

4. Dashboard (Tasks 4 and 5)

The report page contains three KPI cards (Total Accidents, Avg Severity, High Severity %), a map of accidents by city, a column chart by hour, a bar chart by weather, a line chart over time, and three slicers (Day or Night, Severity, Year).



Final dashboard (all slicers set to All)

5. Key findings

- California accounts for about 2 million accidents in the dataset (KPI card, rounded), with an average severity of 2.17 on a 1 to 4 scale; 16.38% of accidents have severity 3 or 4.
- Los Angeles has the most accidents by a large margin, followed by Sacramento and San Diego, which fits the size and traffic of these cities.
- Accidents peak during commuting hours: a first peak around 07:00 - 08:00 and a higher one between 15:00 and 17:00. The lowest counts are in the early morning (02:00 - 04:00).
- Most accidents happen in fair or clear weather, which suggests that traffic volume matters more than bad weather. This is counts only; accident rates would need exposure (traffic) data.
- The monthly trend is low and stable until about 2019, drops sharply in 2020, then rises to a higher level. The drop is probably linked to reduced traffic in 2020; the later increase may partly reflect broader data collection rather than more accidents.
- Slicers update all visuals and the KPI cards. For example, selecting Night reduces Total Accidents to 607K, while average severity (2.16) and the high-severity share (16.07%) stay almost unchanged, and the hourly chart shows only night hours.

6. Limitations

- Results cover California only and are descriptive: they show when and where accidents occur, not why.
- The dataset is built from several data sources, so changes over time can reflect collection coverage as well as real changes.